INDUSTRY PROBLEM STATEMENT

PEN Foundation AI Planner

AI-Powered Foundation Count, Load Analysis & Positioning Engine

Powered by **CDISC Technologies** · Track: Civil Engineering · 42 Hours

HackS'US Edition V · March 6–8, 2025 · Team Size: Up to 4 · Participants: 25–50

01 About the PEN Foundation System

The **Pre-Engineered Nail (PEN) Foundation** is CDISC Technologies' patented structural innovation — a factory-prefabricated foundation system that replaces conventional cast-in-place RCC footings with precision-engineered components installed on-site in a fraction of the time. Unlike conventional shallow foundations designed empirically by a structural engineer for each project, PEN Foundations are **data-driven and load-optimised** — the number of units required, their placement positions, and per-unit load specifications are computationally determinable from structural design inputs and geotechnical data.

Specification Parameter	Value / Detail
Model Reference	CD-PEN-32.3.2.1500
Concrete Node Head	M50 Grade (7,250 psi) precast concrete — IS 456:2000 & IS 10262:2009 compliant; 7-day strength: 41.1 N/mm²
Bearing Nails	4× galvanized steel pipes, Ø32 mm OD × 3.2 mm wall — IS 4923 / IS 3601 / IS 1161; batter angle 51° (standard); optimal at 40° for max vertical capacity
Tip Technology	Tungsten carbide tips for penetration into soil and soft rock — no pre-drilling required
Bolt Assembly	4× M12 galvanized bolts, Grade 4.6 (IS 5624); 100 mm c/c bolt spacing; accommodates structural base plate
Minimum SBC Required	71.8 kN/m²

Minimum Installation Depth	900 mm below ground level
Node Head Footprint	300 mm × 300 mm (plan); suitable for single, double, and triple-node grouped configurations
Certified Field Performance	Field PLT (NIT Calicut): 64.18 kN vs. 33.34 kN (plate) — 1.93× improvement. Clay sites (Kalpetta, Wayanad): 650–725 kN/m² bearing vs. 250–355 kN/m² conventional at 25 mm settlement. Validated by NIT Calicut & UltraTech Lab.
FEA-Validated Behaviour	ANSYS/PLAXIS 3D: Settlement 5.6 mm @ 15 MT (single node, clay); 7 mm @ 30 MT (2-node, clay). Groundwater at surface reduces UBC by 1.66×. Batter angle of 40° yields 3.22× BCR improvement.
Structural Compatibility	Low-rise structures (G+0 to G+3) with MS or RCC framing; compatible with modular net-zero energy homes

Conventional Foundation — Limitations	PEN Foundation — Advantages
Manual load calculations — time-intensive and error-prone	Prefabricated to exact structural specs — zero on-site waste
Foundation positions by intuition — no spatial optimisation	M50 concrete: 41.1 N/mm² @ 7 days — 25–40% higher compressive strength than M30 conventionals
No automated check against SBC; no weak-zone flagging	Field-validated 1.93–2.5× load-bearing improvement over conventional plate foundations
Redesign required if soil varies — no spatial geotechnical awareness	Site-specific design via SBC mapping and GPS soil data integration
No reference learning from past projects	3–5× faster installation vs conventional RCC; semi-skilled labour sufficient

02 The Industry Problem

Today, the engineering workflow for foundation design is **manual, non-scalable, and non-transferable**. When CDISC's PEN Foundation system is specified for a project, there is currently no intelligent tool to determine how many units are needed, where they should be placed, and how loads are distributed — purely from the structural plan and soil data. The five-step conventional workflow illustrates the gap:

Step	Task	Engineering Limitation
1	Manual column load takeoff from drawing	Prone to transcription errors; no parametric feedback loop
2	Manual SBC lookup from PDF soil report	SBC often misread or applied uniformly across non-uniform soil profiles
3	Engineer decides footing count and positions by experience	No optimisation logic; over-engineering is common (waste of material)
4	Spreadsheet calculations — no reference library cross-check	No accumulation of institutional knowledge across projects
5	Process restarts if site soil conditions vary spatially	Zero automated spatial awareness; redesign adds days of delay

That gap — between structural design plan and an optimised, engineer-reviewable PEN Foundation specification — is exactly what this challenge asks participants to close.

03 The Challenge Statement

Build an AI-powered software that:

1. Accepts a structural design plan (PDF/image/DWG) and soil investigation report as primary inputs
2. Parses column grid, span dimensions, slab type, and floor count to compute factored column base loads per IS 875 / IS 456
3. Extracts Safe Bearing Capacity (SBC), soil classification, stratum depth, and water table level from the uploaded geotechnical report via OCR/NLP
4. Determines the number, position, and per-unit load allocation of CDISC PEN Foundation units — matched to the CD-PEN-32.3.2.1500 product specification
5. Optionally auto-fetches live soil data using site GPS coordinates, eliminating manual report upload
6. Validates its output against CDISC's reference design library to quantify accuracy and flag deviations

04 Software Inputs

The solution must accept all four inputs below. Teams may implement these via file upload, form-based entry, or API integration.

#	Input	Description & Engineering Scope	Status
I1	**Architectural / Structural Design Plan**	PDF, JPG/PNG, or DWG. AI must extract: column grid coordinates, beam spans, slab type (flat/ribbed/waffle), number of floors, and occupancy classification. Output: point loads at each column base per IS 875 Part 1 (dead) & Part 2 (live).	**Required**
I2	**Soil Investigation Report**	PDF via OCR/NLP (Tesseract, PyMuPDF, LangChain). Extract: Safe Bearing Capacity (SBC in kN/m²), soil classification (IS 1498), founding stratum depth, water table depth, and any weak/expansive zones. Note: PEN Foundation requires minimum SBC of 71.8 kN/m²; water table at surface reduces effective SBC by ~1.66×.	**Required**
I3	**GPS / Site Coordinates**	Latitude & longitude of construction site. Use to auto-fetch geotechnical data from public databases (USGS, BHDB, GSI, Bhukosh). Even partial return — soil classification or approximate SBC range — earns bonus points. Full automation earns maximum bonus.	**Bonus**
I4	**CDISC Reference Design Library**	Curated dataset of past CDISC projects: structural drawings, SBC values, final foundation counts, X-Y positions, per-unit load allocations. AI must use this as validation context — not substitute it. Provided by CDISC at hackathon start.	**Required**

05 Required Software Outputs

All four outputs below are mandatory. Each must be reviewable by a practising structural engineer — not a black box.

#	Output	Content & Engineering Justification Required	Format
O1	**Foundation Count**	Total PEN units required, with per-unit engineering basis: tributary area, factored load, footing area = Factored Load ÷ SBC, matched CD-PEN model. Must flag if soil SBC is below 71.8 kN/m² threshold and recommend deeper installation or soil treatment.	**Core Output**

O2	**Foundation Position Map**	X-Y coordinate layout or annotated plan showing precise placement of each PEN unit: aligned to column grid, load paths, and SBC zones. Must check for overlap, minimum spacing, and edge distances. Flag positions in weak-zone areas. Single, double, and triple-node groupings must be distinguishable.	**Core Output**
O3	**Load Distribution Report**	Per-foundation breakdown: tributary area (m²) | DL (kN) + LL (kN) | Factored Load = 1.5×(DL+LL) kN | Footing area required (m²) | Soil pressure utilised (kN/m²) | Safety factor. Must flag in red any foundation point where soil bearing pressure exceeds SBC.	**Core Output**
O4	**Reference Design Comparison**	Match submitted design against nearest entry in CDISC reference library (matched by: building footprint area, floor count, SBC range). Output: % deviation in foundation count, positional alignment score, load distribution delta. Flag significant deviations (>15%) for engineer review.	**Core Output**

06 Engineering Logic the AI Must Implement

The following six-step logic chain defines the **minimum computational scope** of a valid solution. All calculations must be traceable and based on IS code provisions.

Step	Engineering Task	What the AI Must Do
S1	**Load Takeoff**	Parse plan to extract column positions, beam spans, slab areas, and floor count. Calculate: DL per slab type (IS 875 Part 1) and LL per occupancy (IS 875 Part 2). Compute Factored Load = 1.5 × (DL + LL) at each column base in kN.
S2	**Soil Data Extraction**	OCR/NLP parse of soil report: extract SBC (kN/m²), soil classification (IS 1498), stratum depth (m), water table depth (m), and weak-zone flags. If GPS provided: auto-query geotechnical databases. Apply 1.66× reduction factor to SBC where water table is at or near surface.
S3	**Foundation Sizing**	For each column point: Required area = Factored Load ÷ SBC. Match to CD-PEN-32.3.2.1500 unit or multi-node grouping (double/triple node per FEA data). Verify minimum SBC ≥ 71.8 kN/m²; recommend micro-concrete fill or under-reamed bulb for marginal sites.
S4	**Foundation Positioning**	Place PEN units at column grid intersections. Enforce: (a) no overlap of soil influence zones (single PEN: 3m×3m×1.5m block), (b) minimum spacing rules, (c) edge/setback distances. Flag positions where SBC drops below threshold; recommend deeper installation or soil improvement.

S5	**Count Optimisation**	Check if adjacent lightly-loaded columns can share a combined double or triple-node PEN unit (soil frame: 4m×4m×4m). Optimise total unit count while maintaining safety margins. Document the count reduction and structural justification for each grouped unit.
S6	**Reference Validation**	Match submitted design to closest CDISC reference entry by: footprint area, floor count, and SBC range. Output: % count deviation, positional alignment score, load delta per foundation. Flag deviations >15% as 'Requires Engineer Review'. Full match within 10% earns highest validation score.

Core Engineering Formula Chain

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Factored Column Load  P  =  1.5 × (DL + LL)  [kN]
Required Footing Area  A  =  P ÷ SBC  [m²]
Soil Bearing Pressure  q  =  P ÷ A_provided  ≤  SBC  [kN/m²]
Safety Factor  FoS  =  SBC ÷ q  ≥  1.0  (flag if < 1.5)
SBC_effective (water table at surface)  =  SBC_dry ÷ 1.66
```

07 Bonus Challenge — Live Soil Data via GPS

+10% Score Bonus

Implement automatic geotechnical data extraction using site GPS coordinates. Entering latitude/longitude should trigger an API query that returns — at minimum — a soil classification or approximate SBC range, eliminating the need for a manually uploaded soil report. Suggested approaches:

- USGS / BHDB / Bhukosh (GSI) — borehole and lithology databases queryable by lat/lon
- Indian Geotechnical Society portals and state PWD soil maps
- Historical borehole interpolation: inverse-distance weighting across surrounding borehole data
- AI-assisted SBC estimation: LLM-based inference from regional soil classification maps + IS 6403 presumed bearing values

Note: Partial implementation (returns soil class / SBC range) qualifies for partial bonus. Full automation of the soil extraction step earns maximum bonus.

08 Judging Criteria

#	Criteria	What Judges Will Evaluate	Weight
1	**Engineering Accuracy**	Correctness of load takeoff (IS 875), factored load computation, footing area calculations, SBC compliance check, and safety factor flagging	**30%**
2	**AI Input Processing**	Quality of design plan parsing (column grid, span detection, floor count) and soil report reading (SBC, stratum depth, water table, weak zones) via OCR/NLP	**25%**
3	**Output Completeness & Clarity**	All four outputs present (O1–O4): count with justification, position map, per-unit load report, reference comparison — legible and actionable for a structural engineer	**20%**
4	**Reference Learning & Validation**	Effectiveness of CDISC reference library usage: match accuracy, deviation quantification, and flag logic for outlier results	**15%**
5	**Bonus: GPS Soil Integration**	Depth and accuracy of automatic geotechnical data retrieval from GPS coordinates — from basic classification return to full SBC estimation	**+10%**

09 Data & Resources Provided by CDISC

The following resources will be distributed to all teams at the start of the hackathon:

Resource	Details
Reference Design Library	Curated past CDISC project dataset: structural drawings, SBC values, PEN unit counts, X-Y foundation positions, and per-unit load values. Teams must use this; external foundation datasets are not permitted.
PEN Foundation Specifications	Technical datasheet for CD-PEN-32.3.2.1500: load capacity envelope, node head footprint (300×300 mm), nail configuration (4× Ø32mm GI pipes, 51° batter, tungsten carbide tips), minimum SBC (71.8 kN/m²), and minimum depth (900 mm).
Sample Soil Reports	3–5 real, anonymised soil investigation reports in PDF — for testing OCR/NLP parsing modules. Include sites with varying SBC, water table conditions, and soil classifications.
Sample Design Plans	3–5 structural floor plans (PDF and image formats): residential G+1, commercial G+3, and industrial single-storey. Cover varied column grids and span configurations.

Engineering Constants	IS 875 loading values: DL per slab type (e.g. 2.0–4.0 kN/m² for RCC slabs), LL per occupancy (1.5 kN/m² residential to 5.0 kN/m² assembly). Reference water table SBC reduction factor: 1.66× (validated by NIT Calicut / PLAXIS 3D study).

10 Constraints & Ground Rules

- The solution must demonstrate at least one complete end-to-end run: design plan input → load calculation → foundation count, load report, and position map output
- Foundation recommendations must reference CDISC PEN Foundation units specifically (CD-PEN-32.3.2.1500) — not generic footing types
- Load calculations must follow IS 456 (structural), IS 875 Part 1 & 2 (loading), and IS 6403 / IS 1888 (bearing capacity) — or equivalent international codes with justification
- Any programming language, AI/ML framework, or API stack is permitted (Python, Node.js, etc.)
- OCR/NLP for soil report parsing may use open-source tools: Tesseract, LangChain, PyMuPDF, Camelot, pdfplumber, etc.
- The CDISC reference design library provided at hackathon start is mandatory; teams may not substitute their own foundation datasets
- Output must be reviewable by a structural engineer — calculations must be visible, traceable, and non-black-box
- Safety factor flagging is mandatory: any foundation point with FoS < 1.5 or bearing pressure exceeding SBC must be explicitly highlighted in the output

11 Impact if Solved

For CDISC Technologies	For the Construction Industry	For the Built Environment
A working prototype becomes a direct commercial asset — a live sales and design tool to demonstrate PEN Foundation suitability, quantity, and estimated cost from any structural plan in real time.	Foundation design time drops from days to minutes. Manual calculation errors are eliminated. PEN Foundation adoption increases because the barrier to specification — complex manual engineering — is removed.	Fewer over-engineered (wasteful) or under-engineered (unsafe) structures. Every project designed with this tool is optimised — less concrete, less embodied carbon, better structural outcomes.

Don't just design the foundation.

Redesign how foundations are designed.

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